

Nifty100 Financial Intelligence Platform

Analyst Guide

Version: 1.0

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Project: Nifty100 Financial Intelligence Platform

Technology Stack: Python | Streamlit | FastAPI | SQLite | Pandas | Plotly

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1. Introduction

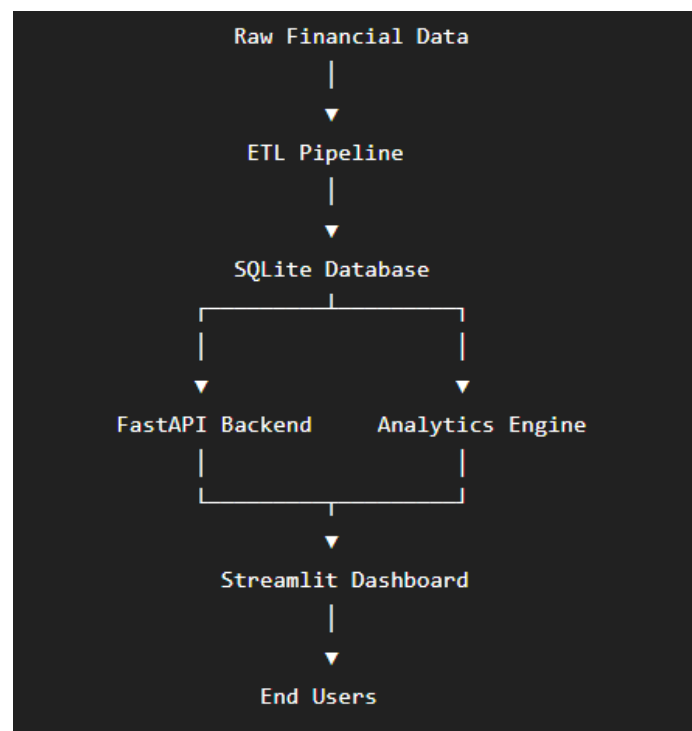
The Nifty100 Financial Intelligence Platform is a financial analytics application developed using Python. The platform provides investors and analysts with interactive dashboards, company financial analysis, stock screening, peer comparison, valuation metrics and downloadable reports. The application integrates financial datasets into a centralized SQLite database and presents insights through Streamlit dashboards and FastAPI endpoints.

2. Project Objectives

The primary objectives of this project are:

- Build a centralized financial analytics platform.
- Analyze financial statements of Nifty100 companies.
- Provide interactive dashboards.
- Compare companies with peers.
- Calculate financial KPIs.
- Generate downloadable reports.
- Provide REST APIs for integration.

3. System Architecture



The architecture separates data ingestion, analytics, APIs, and visualization into independent modules.

4. Technology Stack

Programming Language

- Python 3.10

Libraries

- Pandas
- NumPy
- Plotly
- Streamlit
- FastAPI
- ReportLab
- OpenPyXL

Database

- SQLite

Testing

- PyTest

Code Quality

- Black
- Ruff

5. Installation and Setup

Clone the repository:

```
git clone <repository-url>
```

Move to the project directory:

```
cd nifty100-platform
```

Install dependencies:

```
pip install -r requirements.txt
```

Run the dashboard:

```
streamlit run src/dashboard/app.py
```

Run the API:

```
uvicorn src.api.main:app --reload
```

6. Project Structure

nifty100-platform/

|—— data/

|—— docs/

|—— output/

|—— screenshots/

|—— src/

| |—— analytics/

| |—— api/

| |—— dashboard/

| |—— etl/

| |—— reports/

| |—— scripts/

|—— tests/

|—— README.md

|—— requirements.txt

Each module is organized to improve maintainability and scalability.

7. Database Design

The application stores financial information inside a SQLite database.

Major tables include:

- Companies
- Profit & Loss
- Balance Sheet
- Cash Flow
- Financial Ratios
- Market Capitalization

- Stock Prices
- Sector Information

The database is indexed to improve dashboard performance.

8. Dashboard Modules

Home Dashboard : Provides an overview of the financial platform.

Company Profile : Displays company information and financial metrics.

Stock Screener : Filters companies using predefined financial criteria.

Peer Comparison : Compares selected companies using financial indicators.

Trend Analysis : Displays historical financial trends.

Sector Analysis : Compares companies within the same sector.

Capital Allocation : Visualizes how companies allocate capital.

Annual Reports : Provides downloadable company reports.

Valuation Analytics : Calculates valuation metrics and performance indicators.

9. REST API

The application exposes FastAPI endpoints.

Examples include:

GET /health

GET /api/v1/companies

GET /api/v1/companies/{ticker}

GET /api/v1/screener

GET /api/v1/sectors/{sector}/companies

These APIs allow external systems to consume financial data.

10. Financial Analytics

The analytics module calculates important financial KPIs.

Examples include:

Revenue CAGR

PAT CAGR

EPS CAGR

Debt to Equity

Return on Equity

Operating Margin

Net Profit Margin

Free Cash Flow

CFO Quality Score

CAPEX Intensity

These metrics help evaluate financial health and business performance.

11. Report Generation

The platform generates:

- Company PDF Tearsheet
- Valuation Summary
- Performance Reports
- Financial Reports
- CSV Exports
- Excel Reports

Reports can be shared with analysts and management.

12. Testing

The project uses PyTest for automated testing.

Execute tests:

```
python -m pytest tests -q
```

Current status:

79 Passed

The project also includes API tests, analytics tests, ETL tests, and dashboard-related validations.

13. Performance Optimization

The project includes several optimizations:

- SQLite indexes
- Cached database queries
- Optimized data loading
- Efficient dashboard rendering
- Performance benchmarking

These improvements reduce response time and improve user experience.

14. Troubleshooting

Common issues and solutions:

Dashboard not loading

Verify Streamlit installation and run:

```
streamlit run src/dashboard/app.py
```

Database connection issue

Ensure the SQLite database exists and paths are configured correctly.

Missing dependencies

Run:

```
pip install -r requirements.txt
```

API not starting

Verify FastAPI and Uvicorn installation.

15. Future Enhancements

Possible future improvements include:

- User authentication
- Portfolio tracking
- Live stock market integration
- Machine Learning predictions
- Cloud deployment
- Email report scheduling
- Interactive portfolio analytics

16. Conclusion

The Nifty100 Financial Intelligence Platform provides a complete environment for financial analysis and visualization. By combining ETL pipelines, a centralized SQLite database, FastAPI services, Streamlit dashboards, and automated testing, the project delivers a scalable solution for analyzing Nifty100 companies. The modular architecture also allows future enhancements and easy maintenance.

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